

Ruizhe Zhang

CONTACT INFORMATION	Purdue University Hall of Data Science and AI (DSAI) #3087 West Lafayette, IN 47907, USA	rz Zhang@purdue.edu Homepage, Google scholar
RESEARCH INTERESTS	Theoretical Computer Science, Quantum Computing and Quantum Information, Machine Learning Theory, Classical and Quantum Optimization, Statistical Signal Processing and Inverse Problems, Numerical Linear Algebra.	
EMPLOYMENT	Purdue University <i>Assistant Professor, Department of Computer Science</i>	Fall 2025 - present
	University of California, Berkeley <i>Postdoctoral Scholar, Simons Institute for the Theory of Computing</i> Hosts: Umesh Vazirani and Lin Lin	Fall 2023 - Fall 2025
EDUCATION	The University of Texas at Austin Ph.D. in Computer Science Thesis: <i>Quantum Meets Optimization and Machine Learning</i> Advisor: Dana Moshkovitz	August 2023
	Fudan University B.S. in Computer Science, <i>Honors Class</i>	May 2018
ACADEMIC EXPERIENCE	Institute for Pure & Applied Mathematics at UCLA Position: Departmental Scholar Core Participant of <i>Mathematical and Computational Challenges in Quantum Computing</i>	Fall 2023
PUBLICATIONS (ALPHABETICAL ORDER UNLESS NOT)	[1] Zhiyan Ding, Lin Lin, Yilun Yang, Ruizhe Zhang. Quantum Filtering and Analysis of Multiplicities in Eigenvalue Spectra. <i>PRX Quantum</i>, 7(2), 020318, 2026. [2] Zhiyan Ding, Bowen Li, Lin Lin, Ruizhe Zhang. Polynomial-Time Preparation of Low-Temperature Gibbs States for 2D Toric Code. <i>J. Math. Phys.</i> 67, 052201, 2026. [3] Zhiyan Ding, Haoya Li, Lin Lin, HongKang Ni, Lexing Ying, Ruizhe Zhang. Quantum Multiple Eigenvalue Gaussian filtered Search: an efficient and versatile quantum phase estimation method. In <i>Quantum, Volume 8, Number 1487</i>, 2024. [4] Hongru Yang, Ziyu Jiang, Ruizhe Zhang, Zhangyang Wang, Yingbin Liang. Neural Networks with Sparse Activation Induced by Large Bias: Tighter Analysis with Bias-Generalized NTK. In <i>Journal of Machine Learning Research (JMLR)</i> 25(321): 1-51, 2024. [5] Zhiyan Ding, Ethan N. Epperly, Lin Lin, Ruizhe Zhang. The ESPRIT algorithm under high noise: Optimal error scaling and noisy super-resolution. In <i>Proceedings of the 65th Annual Symposium on Foundations of Computer Science (FOCS)</i>, 2024. [6] Lianke Qin, Zhao Song, Ruizhe Zhang. A General Algorithm for Solving Rank-one Matrix Sensing. In <i>Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)</i>, 2024. [7] Zhao Song, Junze Yin, Lichen Zhang, Ruizhe Zhang. Fast Dynamic Sampling for Determinantal Point Processes. In <i>Proceedings of the 27th International Conference on Artificial</i>	

Intelligence and Statistics (AISTATS), 2024.

[8] Zhao Song, Lichen Zhang, **Ruizhe Zhang**. Training Multi-Layer Over-Parametrized Neural Network in Subquadratic Time. In *Proceedings of the 15th Innovations in Theoretical Computer Science Conference (ITCS)*, 2024.

[9] Josh Alman, Jiehao Liang, Zhao Song, **Ruizhe Zhang**, Danyang Zhuo. Bypass exponential time preprocessing: Fast neural network training via weight-data correlation preprocessing. In *Proceedings of the Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023.

[10] Zhao Song, Baocheng Sun, Omri Weinstein, **Ruizhe Zhang**. Quartic Samples Suffice for Fourier Interpolation. In *Proceedings of the 64th Annual Symposium on Foundations of Computer Science (FOCS)*, 2023.

[11] **Ruizhe Zhang**, Xinzhi Zhang. Hyperbolic Extension of Kadison-Singer Type Results. In *Proceedings of the 50th EATCS International Colloquium on Automata, Languages, and Programming (ICALP)*, 2023.

[12] Guoming Wang, Daniel Stilck França, **Ruizhe Zhang**, Shuchen Zhu, Peter D. Johnson. Quantum algorithm for ground state energy estimation using circuit depth with exponentially improved dependence on precision. In *Quantum, Volume 7, Number 1167*, 2023. In *Quantum Computing Theory in Practice (QCTiP)*, 2023. (Contributed talk) *Author order by contribution

[13] Andrew Childs, Tongyang Li, Jin-Peng Liu, Chunhao Wang, **Ruizhe Zhang**. Quantum Algorithms for Sampling Log-Concave Distributions and Estimating Normalizing Constants. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*, 2022. In *26th Annual Conference on Quantum Information Processing (QIP)*, 2023. (Contributed talk)

[14] Tongyang Li, **Ruizhe Zhang**. Quantum Speedups of Optimizing Approximately Convex Functions with Applications to Logarithmic Regret Stochastic Convex Bandits. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Yichuan Deng, Zhao Song, Omri Weinstein, **Ruizhe Zhang**. Fast Distance Oracles for Any Symmetric Norm. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*, 2022.

[16] Baihe Huang, Shunhua Jiang, Zhao Song, Runzhou Tao, **Ruizhe Zhang**. Solving SDP Faster: A Robust IPM Framework and Efficient Implementation. In *Proceedings of the 63rd Annual Symposium on Foundations of Computer Science (FOCS)*, 2022.

[17] **Ruizhe Zhang**, Guoming Wang, Peter Johnson. Computing Ground State Properties with Early Fault-Tolerant Quantum Computers. In *Quantum, Volume 6, Number 761*, 2022. *Author order by contribution

[18] Jason Gaitonde, Max Hopkins, Tali Kaufman, Shachar Lovett, **Ruizhe Zhang**. Eigenstripping, Spectral Decay, and Edge-Expansion on Posets. In *Proceedings of the 26th International Conference on Randomization and Computation (RANDOM)*, 2022.

[19] Zhao Song, **Ruizhe Zhang**. Hyperbolic Concentration, Anti-concentration, and Discrepancy. In *Proceedings of the 26th International Conference on Randomization and Computation (RANDOM)*, 2022.

[20] Sitan Chen, Zhao Song, Runzhou Tao, **Ruizhe Zhang**. Symmetric Sparse Boolean Matrix Factorization and Applications. In *Proceedings of the 13th Innovations in Theoretical*

Computer Science Conference (ITCS), 2022.

[21] Nai-Hui Chia, Chi-Ning Chou, Jiayu Zhang, **Ruizhe Zhang**. Quantum Meets Minimum Circuit Size Problem. In *Proceedings of the 13th Innovations in Theoretical Computer Science Conference (ITCS)*, 2022.

[22] Zhao Song, Shuo Yang, **Ruizhe Zhang**. Does Preprocessing Help Training Over-Parameterized Neural Networks? In *Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS)*, 2021.

[23] Yuxuan Zhang, **Ruizhe Zhang**, Andrew C. Potter. QED driven QAOA for network-flow optimization. In *Quantum, Volume 5, Number 510*, 2021.

[24] Scott Aaronson, Jiahui Liu, Qipeng Liu, Mark Zhandry, **Ruizhe Zhang**. New Approaches for Quantum Copy-Protection. In *Proceedings of the 41st Annual International Cryptology Conference (CRYPTO)*, 2021.

[25] Scott Aaronson, Nai-Hui Chia, Han-Hsuan Lin, Chunhao Wang, **Ruizhe Zhang**. On the Quantum Complexity of Closest Pair and Related Problems. In *Proceedings of the 35th Computational Complexity Conference (CCC)*, 2020.

MANUSCRIPTS

[1] **Ruizhe Zhang**. Optimal Extrapolation Bounds for Sparse Fourier Sums. *arXiv preprint arXiv:2607.10501*, 2026.

[2] Hongrui Chen, Zhiyan Ding, **Ruizhe Zhang**. Overcoming the Lamb Shift in System-Bath Models via KMS Detailed Balance: High-Accuracy Thermalization with Time-Bounded Interactions. *arXiv preprint arXiv:2604.15616*, 2026.

[3] Yeqi Gao, Zhao Song, **Ruizhe Zhang**. Quantum Speedup for Spectral Approximation of Kronecker Products. *arXiv preprint arXiv:2402.07027*, 2024.

[4] Zhao Song, Junze Yin, **Ruizhe Zhang**. Revisiting Quantum Algorithms for Linear Regressions: Quadratic Speedups without Data-Dependent Parameters. *arXiv preprint arXiv:2311.14823*, 2023.

[5] Yeqi Gao, Zhao Song, Xin Yang, **Ruizhe Zhang**. Fast quantum algorithm for attention computation. *arXiv preprint arXiv:2307.08045*, 2023.

[6] Yichuan Deng, Zhao Song, Lichen Zhang, **Ruizhe Zhang**. Efficient Algorithm for Solving Hyperbolic Programs. *arXiv preprint arXiv:2306.07587*, 2023.

[7] Baihe Huang, Shunhua Jiang, Zhao Song, Runzhou Tao, **Ruizhe Zhang**. A Faster Quantum Algorithm for Semidefinite Programming via Robust IPM Framework. *arXiv preprint arXiv:2207.11154*, 2022.

[8] Yeqi Gao, Zhao Song, Baocheng Sun, Omri Weinstein, **Ruizhe Zhang**. Improved Reconstruction for Fourier-Sparse Signals. *arXiv preprint arXiv:2205.00658*, 2022.

[9] Baihe Huang, Zhao Song, Omri Weinstein, Hengjie Zhang, **Ruizhe Zhang**. A Dynamic Fast Gaussian Transform. *arXiv preprint arXiv:2202.12329*, 2022.

[10] Baihe Huang, Zhao Song, Runzhou Tao, **Ruizhe Zhang**, Danyang Zhuo. InstaHide's Sample Complexity When Mixing Two Private Images. *arXiv preprint arXiv:2011.11877*, 2020.

TALKS

Overcoming the Lamb shift: high-accuracy quantum Gibbs sampling with system-bath

interactions

- Invited talk at MQC Entanglement 2026 Jul. 2026

Quantum Filtering and Analysis of Multiplicities in Eigenvalue Spectra

- Invited talk at the 2nd reunion of “Mathematical and Computational Challenges in Quantum Computing” Jun. 2026

Recent Progress in Quantum Eigenvalue Problems

- Invited talk at 2025 SIAM NNP Conference Nov. 2025
- Invited talk at Purdue Quantum AI (PQAI) 2025 workshop Oct. 2025

Recent Progress in Post-Kitaev Quantum Phase Estimation

- Invited talk at the 1st reunion of “Mathematical and Computational Challenges in Quantum Computing” Jun. 2025

Polynomial-Time Preparation of Low-Temperature Gibbs States for 2D Toric Code

- Invited talk at Quantum Pod Seminar at UC Berkeley Nov. 2024

The ESPRIT algorithm under high noise: Optimal error scaling and noisy super-resolution

- The 65th Annual Symposium on Foundations of Computer Science (FOCS 2024) Oct. 2024

Early Fault-Tolerant Approaches for Hamiltonian Eigenvalues Estimation

- Invited talk at Quantum Meets Mathematics Seminar at Ohio State University Oct. 2024

Bridging Quantum and Classical Fast Algorithms for Optimization and Machine Learning

- Invited talk at Computer Science Colloquium at Rice University Nov. 2023

Quantum Speedups of Continuous Sampling and Optimization Problems

- Invited talk at NSF Post-Quantum AI Workshop Apr. 2024
- Invited talk at IPAM CQC 2023 Workshop I Oct. 2023
- Invited talk at George Mason University Theory Seminar Apr. 2023
- Invited talk at MIT Quantum Information Processing Seminar Jan. 2023

Quartic Samples Suffice for Fourier Interpolation

- The 64th Annual Symposium on Foundations of Computer Science (FOCS 2023) Nov. 2023

Quantum Algorithm for Ground State Energy Estimation Using Circuit Depth With Exponentially Improved Dependence on Precision

- Contributed talk at APS March Meeting 2023 Mar. 2023

Solving SDP Faster: A Robust IPM Framework and Efficient Implementation

- Invited talk at UT Austin Theory Seminar Apr. 2023
- The 63rd Annual Symposium on Foundations of Computer Science (FOCS 2022) Nov. 2022

Ground State Energy and Property Estimation with Low Quantum Circuit Depth

- Invited talk at QUARK Lab at Peking University Oct. 2022

Hyperbolic Concentration, Anti-concentration, and Discrepancy

	– The 26th International Conference on Randomization and Computation (RANDOM 2022) Sep. 2022
	On the Quantum Fine-Grained Complexity of Closest Pair and Related Problems – Invited talk at University of Washington Theory Seminar May 2022
	Computing Ground State Properties with Early Fault-Tolerant Quantum Computers – Contributed talk at APS March Meeting 2022 Mar. 2022
	Symmetric Sparse Boolean Matrix Factorization and Applications – The 13th Innovations in Theoretical Computer Science Conference (ITCS 2022) Feb. 2022
	Quantum Speedup for Annealing and Sampling – Invited talk at Simons Institute reading group Dec. 2021
	Quantum Meets the Minimum Circuit Size Problem – Second Kyoto Workshop on Quantum Information, Computation, and Foundation (QICF 2021) Sep. 2021
	On the Quantum Complexity of Closest Pair and Related Problems – Computational Complexity Conference (CCC 2020) Sep. 2020 – The 15th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2020) Aug. 2020
STUDENTS	Ph.D. students supervised – Deming Chu (2026 - present) Undergraduate Students supervised – Alexandre Apollinaire Sauquet (2026)
TEACHING	At Purdue University – CS 585 - Theoretical Computer Science Toolkit Spring 2026 – CS 593 - Algorithms for Data Science Fall 2025 At UT Austin – TA for CS395T, Coding Theory (graduate class) Spring 2022 – TA for CS388R, Randomized Algorithms (graduate class) Fall 2021 – TA for CS388T, Theory of Computation (graduate class) Spring 2020 At Fudan University – TA for Data Structure (undergraduate class) Fall 2016
SERVICES	Program committees: – TQC 2025, PQAI 2025, NeurIPS 2026 (Area Chair) Journal refereeing: – SIAM Journal on Computing, SIAM Journal on Optimization, Quantum, PRX Quantum, Physical Review Letter, Physical Review A, Physical Review Research, Transactions on Information Theory, Advances in Engineering Software, npj Quantum Information, ACM Journal on Autonomous Transportation Systems, Journal of Machine Learning Research

Conference refereeing:

- ESA 2020, Eurocrypt 2021, QIP 2021, AQIS 2021, QIP 2022, STOC 2022, ICML 2022, SODA 2023, QIP 2023, AAAI 2023, STACS 2023, STOC 2023, TQC 2023, CRYPTO 2023, FOCS 2023, NeurIPS 2023, QIP 2023, AAAI 2024, ICLR 2024, AISTATS 2024, ICML 2024, QIP 2024, QCTIP 2024, ICALP 2024, TQC 2024, NeurIPS 2024, AAAI 2025, ICLR 2025, AISTATS 2025, ICML 2025, SODA 2025, AAAI 2026, ITCS 2026, AISTATS 2026, QIP 2026, TQC 2026, ICML 2026, FOCS 2026, SODA 2027

Grant refereeing:

- National Science Foundation, 2025